

Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

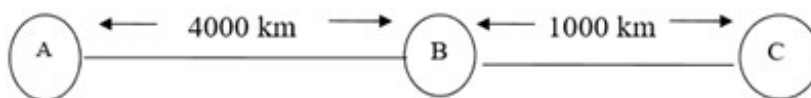
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: ECE 3123/ECE_3123 COMMUNICATION NETWORKS
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) The frames of 1Kb each are generated at node A and sent to node C through node B. The links (4)
between A and B, B and C are full-duplex, error-free.



The data rate between A and B is 100 kbps. The propagation delay is 5 μ sec/km for both lines; ACK frames are separate frames of negligible length. Between A and B, a sliding window protocol is used, with a window size of 3.

Between B and C, stop-and-wait protocol is used.

Find the data rate of B to C link to avoid buffer overflow at node B.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 1B) Twenty digital sources are multiplexed using synchronous TDM. Out of which five are with a bit rate (3)
of 120 kbps, ten with a bit rate of 60 kbps, and the last five with a bit rate of 110 kbps. Design a
synchronous TDM for these sources (The unit of data is 1 bit) and find: a) The output frame rate b)
The output data rate c) Efficiency of the system.

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 1C) Draw the ATM reference model and explain the functions of each layer. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) Draw the flow diagram of CSMA/CD and explain its features. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Consider a 200m, 4Mbps token ring containing 20 stations, each transmitting with equal priority. (3)
Suppose no station is allowed to transmit more than 5000 bytes before giving up the token. Once a
station gives up a token how long will it take (in the worst case) for that station to get the token
again? Assume Speed of signal is 200 meters/ μ sec. Assume frame processing time at each station
is 2.5- μ sec.

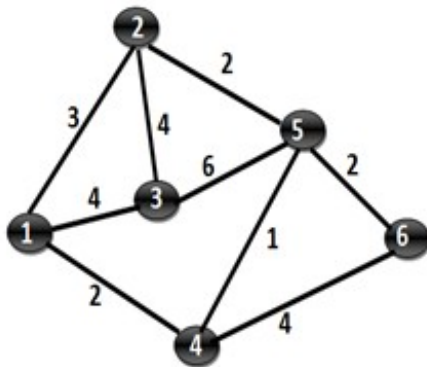
Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 2C) Draw the FDDI frame structure and explain each field in detail. (3)

- 3A) If the IPv4 packet is received with the following header, verify for errors if any, using an appropriate error control technique. (4)

0x4	0x5	0x00	(44) ₁₀	
9D08			2	0
(128) ₁₀		0x06	8BFF	
128.143.137.144				
128.143.71.21				

- 3B) Use Dijkstra's algorithm to find the shortest distance between nodes '1' and '6'. (3)

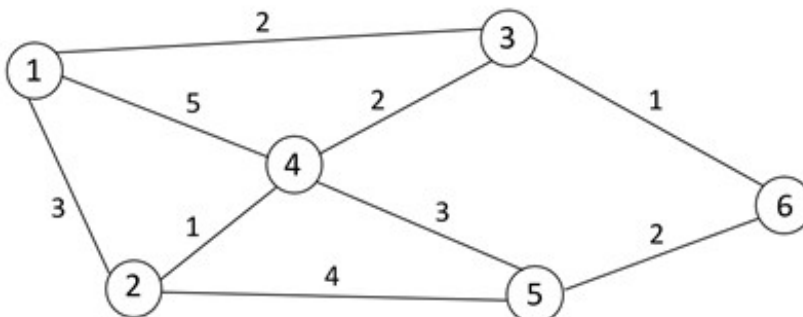


NOTE: Tabulate the routing updates.

- 3C) i) MAHE network having a capacity of 100Mb of data is arriving at the rate of 25 Mbps for 4sec. The Token arrival rate is 2 Mbps and the capacity of bucket is 50Mb with maximum output rate 100 Mbps. Find the burst Length and total output time. (3)
- ii) Discuss the congestion control techniques in datagram subnets.

- 4A) MAHE is granted a block of IPv4 addresses with one of the addresses is 14.24.74.80/24. MAHE needs to allocate 120 addresses to MIT, 60 addresses to KMC, and 10 addresses to SOIS. Design the subblocks and find out how many addresses are still available if any, after these allocations. (4)

- 4B) Using Bellman Ford routing algorithm, find the shortest distances at all the nodes before and after link failure between nodes 3 and 6, if the destination node is 6. (3)



NOTE: Tabulate the routing updates.

4C) Draw the IPv4 header format and explain the functions of each field. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

5A) Draw the state transition diagram for connection management in TCP. Explain the state sequence at client and server. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5B) Explain the functions of TCP timers. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) With suitable diagrams, explain the functions of FTP client and server. Also mention the details of connections in FTP. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_3123:COMMUNICATION NETWORKS

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	3
3	3	CO 3	14
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_3123:COMMUNICATION NETWORKS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	16(32%)
3	Application	3	10(20%)
4	Analysis	0	0(0%)
5	Create	7	24(48%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)